

STAT 5810/6810 Introduction to Statistical Computing

Instructions for Homework 2

by

Jürgen Symanzik

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1 L^AT_EX Installation and Usage

1.1 L^AT_EX Installation

L^AT_EX has been installed on many computers at Utah State University (USU). If you need to install L^AT_EX on your own computer, google for it. For PCs, a frequently used version of L^AT_EX is MikTeX which can be obtained from <http://miktex.org/>. Follow the download and installation instructions.

1.2 L^AT_EX Usage

There exist many support tools that simplify the usage of L^AT_EX. I use *TeXworks* from <http://www.tug.org/texworks/>. Depending on your L^AT_EX installation, *TeXworks* may be installed automatically as part of the L^AT_EX installation. If not, follow the download and installation instructions. Other tools that simplify the use and translation of L^AT_EX documents also exists. It is up to you to explore what is available. Talk to some of the other graduate students who use L^AT_EX for their MS and PhD work and let them explain which support tools (if any) they use for their work with L^AT_EX.

Most of these tools allow you to translate a L^AT_EX document (a file with extension `.tex`) via the click of one (or a few) buttons.

If you can't decide which support tool to install, you will have to translate your L^AT_EX files manually. Change your working directory to the directory where the main `.tex` file is located. Then type `pdflatex filename` (assuming your main file is called `filename.tex`). This will create a file called `filename.pdf` plus several support files. These files will be used in future steps to create the table of context, the reference list, an index section, and so on.

Just to be on the safe side that everything gets properly updated, you should run `pdflatex filename` three times before you finally submit a document — or otherwise, it may happen that a last minute shift of a section heading to a new page may not be reflected in the final table of contents.

If you installed a support tool, this tool almost certainly will know how many times it needs to repeat the `pdflatex filename` sequence (and any other steps that may be needed).

1.3 Images in L^AT_EX

1.3.1 pdf, jpg, and png Format

L^AT_EX supports most major types of images, but those cannot be mixed in arbitrary ways. `pdflatex`, as described above, allows you to incorporate images in pdf, jpg, and png format.

1.3.2 ps and eps Format

If you happen to have images in ps or eps format, then you have to translate your document via `latex filename`. This will create a file called `filename.dvi` plus several support files. You can view this file directly (check the help pages of your installation), or you can translate it via `dvips filename` or `dvi2ps filename` into a ps document (again, check the help pages of your installation). You can view this ps file directly with a postscript viewer or translate it further into pdf with Adobe Acrobat. In any case, working with ps & eps images is more complicated, but such images usually are of a higher quality than jpg images in particular.

1.3.3 tiff Format

L^AT_EX does not support tiff images, although many publishers request tiff images. In such a case, you need to create a working copy of your image (e.g., in jpg format) and use that for your local L^AT_EX translation. Once you have submitted your L^AT_EX source files to the publisher, they will replace the working copy of your image with the high-resolution tiff image on their side at the final production stage of a book chapter or journal article.

1.3.4 Inclusion of Images

Below is an example that shows how to include a pdf image. L^AT_EX allows you to determine where to place a figure (here at the bottom of the page), adjust the width (or height) of a figure (here, 99% of the maximum textwidth), and later reference this figure via `Figure~\ref{fig_CCmap}`:

```

\begin{figure}[b]
\includegraphics[width=0.99\textwidth]{fig_CCmap.pdf}
\caption{\label{fig_CCmap}
CCmaps, based on data from the USDA--NASS web page,
related to soybean production in the United States.
The plot shows the dependent variable production [top slider] that is
conditioned on the two independent variables acreage [bottom slider]
and yield [right slider].
}
\end{figure}

```

Whenever you include figures (and tables), create meaningful captions! Also, keep in mind that each figure must be referenced at least once in the text. Figures (and tables) **must** be numbered in the order they are referenced in the text. You can't refer to Fig. 2 before you refer to Fig. 1. In such a case, you must change the order of your figures in your document.

2 Homework 2

In most cases, we start with a $\text{\LaTeX}$ template. There exists templates for MS reports, MS theses, and PhD dissertations at USU. Check with other graduate students for good templates. Also, many journals, major publishers, or organizers for conferences provide $\text{\LaTeX}$ templates for their publications. Authors basically are asked to keep the overall formatting in place and just make changes to the template, e.g., filling in author names, title, abstract, and, of course, the actual text.

For Homework 2, you have to do the same, i.e., start with this template, fill in your name, and your answers, but keep the overall formatting, title page, and so on. During the next few months, we will extend this template with additional $\text{\LaTeX}$ features.

In particular, for this homework, incorporate the answers in your .R file as follows:

Your answers to Homework 2 should be incorporated in a verbatim environment (see the .tex source file for details).

See w h a t this environment does

1

2

3

4

5

End (of verbatim).

Just as a comparison, below are the previous seven lines, now outside a verbatim environment:

See w h a t this environment does 1 2 3 4 5 End (of verbatim).

All you need to submit for this homework is the final resulting pdf file.

3 Getting Help

There exist numerous sources for help with L^AT_EX:

1. The L^AT_EX Wikibook at <http://en.wikibooks.org/wiki/LaTeX>.
2. The Wikipedia page of L^AT_EX at <http://en.wikipedia.org/wiki/LaTeX> with many excellent references (online and books).
3. Information from the TeX Users Group (TUG) at <http://www.tug.org/begin.html>.

As mentioned above, many of the other graduate students at USU are using L^AT_EX. Help may be available at the next desk. Otherwise, ask some of the other students in class, use Canvas to ask a question, or directly contact me!

Finally, do not forget to google! I usually include the word `latex` in my google query when I search for a particular L^AT_EX command.

Good luck!!!